

8.3 Homogeneous and universal graphs

Unlike finite graphs, infinite graphs offer the possibility to represent an entire graph property $\mathcal{P}$ by just one specimen, a single graph that contains all the graphs in $\mathcal{P}$ up to some fixed cardinality. Such graphs are called ‘universal’ for this property.

universal

More precisely, if $\leq$ is a graph relation (such as the minor, topological minor, subgraph, or induced subgraph relation up to isomorphism), we call a countable graph G^* *universal* in $\mathcal{P}$ (for $\leq$) if $G^* \in \mathcal{P}$ and $G \leq G^*$ for every countable graph $G \in \mathcal{P}$.

Is there a graph that is universal in the class of *all* countable graphs? Suppose a graph R has the following property:

*Whenever U and W are disjoint finite sets of vertices in R ,
there exists a vertex $v \in R - U - W$ that is adjacent in R
to all the vertices in U but to none in W .* (*)

Then R is universal even for the strongest of all graph relations, the induced subgraph relation. Indeed, in order to embed a given countable graph G in R we just map its vertices $v_1, v_2, \dots$ to R inductively, making sure that v_n gets mapped to a vertex $v \in R$ adjacent to the images of all the neighbours of v_n in $G[v_1, \dots, v_n]$ but not adjacent to the image of any non-neighbour of v_n in $G[v_1, \dots, v_n]$. Clearly, this map is an isomorphism between G and the subgraph of R induced by its image.

[11.3.5]
 R

Theorem 8.3.1. (Erdős & Rényi 1963)

There exists a unique countable graph R with property ().*

Proof. To prove existence, we construct a graph R with property (*) inductively. Let $R_0 := K^1$. For all $n \in \mathbb{N}$, let R_{n+1} be obtained from R_n by adding for every set $U \subseteq V(R_n)$ a new vertex v joined to all the vertices in U but to none outside U . (In particular, the new vertices form an independent set in R_{n+1} .) Clearly $R := \bigcup_{n \in \mathbb{N}} R_n$ has property (*).

To prove uniqueness, let $R = (V, E)$ and $R' = (V', E')$ be two graphs with property (*), each given with a fixed vertex enumeration. We construct a bijection $\varphi: V \rightarrow V'$ in an infinite sequence of steps, defining $\varphi(v)$ for one new vertex $v \in V$ at each step.

At every odd step we look at the first vertex v in the enumeration of V for which $\varphi(v)$ has not yet been defined. Let U be the set of those of its neighbours u in R for which $\varphi(u)$ has already been defined. This is a finite set. Using (*) for R' , find a vertex $v' \in V'$ outside the image of φ (which is a finite set), so that v' is adjacent in R' to all the vertices in $\varphi(U)$ but to no other vertex in the image of φ . Put $\varphi(v) := v'$.

At even steps in the definition process we do the same thing with the roles of R and R' interchanged: we look at the first vertex v' in the enumeration of V' that does not yet lie in the image of φ , and set

$\varphi(v) = v'$ for a new vertex v that matches the adjacencies and non-adjacencies of v' among the vertices for which φ (resp. φ^{-1}) has already been defined.

By our minimum choices of v and v' , the bijection gets defined on all of V and all of V' , and it is clearly an isomorphism. $\square$

The graph R in Theorem 8.3.1 is usually called the *Rado graph*, named after Richard Rado who gave one of its earliest explicit definitions. The method of constructing a bijection in alternating steps, as in the uniqueness part of the proof, is known as the *back-and-forth* technique. *Rado graph*

The Rado graph R is unique in another rather fascinating respect. We shall hear more about this in Chapter 11.3, but in a nutshell it is the following. If we generate a countably infinite *random* graph by admitting its pairs of vertices as edges independently with some fixed positive probability $p \in (0, 1)$, then with probability 1 the resulting graph has property $(*)$, and is hence isomorphic to R ! In the context of infinite graphs, the Rado graph is therefore also called *the* (countably infinite) *random graph*. *infinite random graph*

As one would expect of a random graph, the Rado graph shows a high degree of uniformity. One aspect of this is its resilience against small changes: the deletion of finitely many vertices or edges, and similar local changes, leave it ‘unchanged’ and result in just another copy of R (Exercise 54).

The following rather extreme aspect of uniformity, however, is still surprising: no matter how we partition the vertex set of R into two parts, at least one of the parts will induce another isomorphic copy of R . Trivial examples aside, the Rado graph is the only countable graph with this property, and hence unique in yet another respect:

Proposition 8.3.2. *The Rado graph is the unique countable graph G other than $K^{\aleph_0}$ and $\overline{K}^{\aleph_0}$ such that, no matter how $V(G)$ is partitioned into two parts, one of the parts induces an isomorphic copy of G .*

Proof. We first show that the Rado graph R has the partition property. Let $\{V_1, V_2\}$ be a partition of $V(R)$. If $(*)$ fails in both $R[V_1]$ and $R[V_2]$, say for sets U_1, W_1 and U_2, W_2 , respectively, then $(*)$ fails for $U = U_1 \cup U_2$ and $W = W_1 \cup W_2$ in R , a contradiction.

To show uniqueness, let $G = (V, E)$ be a countable graph with the partition property. Let V_1 be its set of isolated vertices, and V_2 the rest. If $V_1 \neq \emptyset$ then $G \not\cong G[V_2]$, since G has isolated vertices but $G[V_2]$ does not. Hence $G = G[V_1] \cong K^{\aleph_0}$. Similarly, if G has a vertex adjacent to all other vertices, then $G = K^{\aleph_0}$.

Assume now that G has no isolated vertex and no vertex joined to all other vertices. If G is not the Rado graph then there are sets U, W for which $(*)$ fails in G ; choose these with $|U \cup W|$ minimum.

Assume first that $U \neq \emptyset$, and pick $u \in U$. Let V_1 consist of u and all vertices outside $U \cup W$ that are not adjacent to u , and let V_2 contain the remaining vertices. As u is isolated in $G[V_1]$, we have $G \not\cong G[V_1]$ and hence $G \cong G[V_2]$. By the minimality of $|U \cup W|$, there is a vertex $v \in G[V_2] - U - W$ that is adjacent to every vertex in $U \setminus \{u\}$ and to none in W . But v is also adjacent to u , because it lies in V_2 . So U , W and v satisfy $(*)$ for G , contrary to assumption.

Finally, assume that $U = \emptyset$. Then $W \neq \emptyset$. Pick $w \in W$, and consider the partition $\{V_1, V_2\}$ of V where V_1 consists of w and all its neighbours outside W . As before, $G \not\cong G[V_1]$ and hence $G \cong G[V_2]$. Therefore U and $W \setminus \{w\}$ satisfy $(*)$ in $G[V_2]$, with $v \in V_2 \setminus W$ say, and then U, W, v satisfy $(*)$ in G . $\square$

Another indication of the high degree of uniformity in the structure of the Rado graph is its large automorphism group. For example, R is easily seen to be *vertex-transitive*: given any two vertices x and y , there is an automorphism of R mapping x to y .

homoge-
neous

In fact, much more is true: using the back-and-forth technique, one can easily show that the Rado graph is *homogeneous*: every isomorphism between two finite induced subgraphs can be extended to an automorphism of the entire graph (Exercise 56).

R^r

Which other countable graphs are homogeneous? The complete graph $K^{\aleph_0}$ and its complement are again obvious examples. Moreover, for every integer $r \geq 3$ there is a homogeneous K^r -free graph R^r , constructed as follows. Let $R_0^r := K^1$, and let R_{n+1}^r be obtained from R_n^r by joining, for every induced subgraph $H \not\supseteq K^{r-1}$ of R_n^r , a new vertex v_H to every vertex in H . Then let $R^r := \bigcup_{n \in \mathbb{N}} R_n^r$. Clearly, as the new vertices v_H of R_{n+1}^r are independent, there is no K^r in R_{n+1}^r if there was none in R_n^r , so $R^r \not\supseteq K^r$ by induction on n . Just like the Rado graph, R^r is clearly universal among the K^r -free countable graphs, and by the back-and-forth argument from the proof of Theorem 8.3.1 it is easily seen to be homogeneous.

By the following deep theorem of Lachlan and Woodrow, the countable homogeneous graphs we have seen so far are essentially all:

Theorem 8.3.3. (Lachlan & Woodrow 1980)

Every countably infinite homogeneous graph is one of the following:

- *a disjoint union of complete graphs of the same order, or the complement of such a graph;*
- *the graph R^r or its complement, for some $r \geq 3$;*
- *the Rado graph R .*

To conclude this section, let us return to our original problem: for which graph properties is there a graph that is universal with this property? Most investigations into this problem have addressed it from a

more general model-theoretic point of view, and have therefore been based on the strongest of all graph relations, the induced subgraph relation. As a consequence, most of these results are negative; see the notes.

From a graph-theoretic point of view, it seems more promising to look instead for universal graphs for the weaker subgraph relation, or even the topological minor or minor relation. For example, while there is no universal planar graph for subgraphs or induced subgraphs, there is one for minors:

Theorem 8.3.4. (Diestel & Kühn 1999)

There exists a universal planar graph for the minor relation.

This remained the only result about universal graphs for the minor relation for over 20 years, until Georgakopoulos found $\preceq$ -universal graphs in $\text{Forb}_{\preceq}(X) = \{G \mid X \not\preceq G\}$ for $X = K^5$ and $X = K_{3,3}$, and proved there is none for $X = K^{\aleph_0}$.

8.4 Connectivity and matching

In this section we look at infinite versions of Menger's theorem and of the matching theorems from Chapter 2. This area of infinite graph theory is one of its best developed fields, with several deep results. One of these, however, stands out among the rest: a version of Menger's theorem that had been conjectured by Erdős decades ago, and was proved only fairly recently by Aharoni and Berger. The techniques developed for its proof inspired, over the years, much of the theory in this area.

Before we turn to this result, however, let us take a brief look at edge-connectivity. Recall from Section 8.1 that in an infinitely edge-connected countable graph we can easily find infinitely many edge-disjoint spanning trees. Can we still find such trees when the graph is uncountable? We can, but this is not quite as easy to prove (Exercise 62).

The following deep theorem of Laviolette reduces the above problem to its countable case – as it does for many other problems involving edge-connectivity. Let $\mathcal{H}$ be a set of countable graphs forming an edge-decomposition of an arbitrary graph G . Call this decomposition *bond-faithful* if every countable bond of G is contained in some $H \in \mathcal{H}$ and every finite bond of any $H \in \mathcal{H}$ is a bond also of G . Note that the finite bonds of G will be bonds of the $H \in \mathcal{H}$ that contain them. (Why?)

Theorem 8.4.1. (Laviolette 2005)

Every graph has a bond-faithful decomposition into countable graphs.

We shall not be able to prove Laviolette's theorem here. But let us illustrate its power in reducing problems to their countable case by deducing an early classic from the theory of infinite graphs.